COMP2300 Set 7 Practice Exam

Topics: Pipelining, Hazards, Forwarding, Stalls, Branch Prediction, Superscalar

Time: 120 minutes **Total:** 100 marks

Instructions

- Answer in English.
- Part A is multiple choice.
- Part B contains pipeline calculations and hazard reasoning.
- Part C contains adapted past-exam style questions with source tags.
- Assume a five-stage ARM pipeline unless stated otherwise.

Part A: Multiple Choice (30 marks, 3 marks each)

Choose exactly one option for each question.

A1. The primary benefit of pipelining is:

(A) Lowering ISA complexity
(B) Improving throughput by overlapping instruction stages
(C) Eliminating branch instructions
(D) Guaranteeing CPI = 1 on all workloads

A2. Which statement is true about latency in a pipelined CPU?

(A) It always decreases for each instruction
(B) It may stay similar or increase because of pipeline-register overhead
(C) It becomes zero after the pipeline fills
(D) It is equal to IPC

A3. A read-after-write hazard is a:

(A) Structural hazard
(B) Control hazard
(C) Data hazard
(D) Cache-coherence hazard

A4. Forwarding helps because it:

(A) Replaces the register file

(B) Sends a result directly from a later pipeline stage to an earlier consumer

(C) Eliminates all control hazards

(D) Prevents all load-use stalls

A5. A load-use hazard often still needs a stall because:

(A) The ALU cannot add two operands in one cycle

(B) The load result becomes available only after the memory stage

(C) The branch predictor is too slow

(D) The decode stage cannot read registers

A6. In the lecture design, a bubble is:

(A) A branch target buffer entry

(B) An empty operation inserted into the pipeline

(C) A renamed register

(D) The same thing as forwarding

A7. Which structure primarily stores branch target addresses?

(A) BHT

(B) BTB

(C) ROB

(D) IQ

A8. A one-bit predictor is weak on loop termination because it:

(A) Cannot distinguish branch PC values

(B) Has no branch target address

(C) Changes direction too easily after a single unusual outcome

(D) Requires two pipeline stages to update

A9. A Smith predictor uses:

(A) 2-bit saturating counters

(B) Floating-point scores

(C) A stack of branch PCs

(D) Only compile-time information

A10. A 2-way superscalar processor ideally can achieve:

(A) IPC up to 0.5

(B) IPC up to 1

(C) IPC up to 2

(D) IPC up to 5

Part B: Computational Problems (50 marks)

B1. Ideal pipeline timing (8 marks)
A five-stage single-issue pipeline executes 20 independent instructions with no hazards.

(a) How many cycles are needed if the pipeline starts empty?
(b) What is the CPI for this run?
(c) What is the IPC for this run?

B2. Forwarding or stall? (8 marks)
Consider the instruction sequence:

```
i1: ADD R1, R2, R3
i2: SUB R4, R1, #1
i3: LDR R5, [R6, #0]
i4: ADD R7, R5, R8
```

(a) Which pair can be handled by forwarding without any stall?
(b) Which pair is a load-use hazard that still requires a stall?
(c) Briefly explain why.

B3. Load-use stall accounting (8 marks)
Assume a five-stage pipeline with forwarding and hazard detection. A program executes 40 instructions, and 6 of them cause exactly one load-use stall each.

(a) How many total cycles are needed if the ideal no-hazard execution would have taken 44 cycles?
(b) What is the effective CPI?

B4. Branch penalty comparison (8 marks)
A branch is taken 30 times in a program.

(a) If the branch misprediction penalty is 4 cycles, how many wasted cycles occur for 30 mispredictions?
(b) If early branch resolution reduces the penalty to 2 cycles, how many cycles are saved over those same 30 mispredictions?

B5. One-bit and two-bit prediction (8 marks)
Consider the branch outcome sequence:

$$T,\ T,\ T,\ T,\ N$$

(a) If a one-bit predictor starts in state `Taken`, how many predictions are correct?
(b) If a two-bit Smith predictor starts in state `Strongly Taken`, what final state does it end in after the sequence?

B6. Superscalar IPC (10 marks)
A 2-way superscalar processor executes a 24-instruction program in 15 cycles.

(a) Compute the achieved IPC.
(b) Compute the achieved CPI.
(c) Briefly explain why the achieved IPC may be lower than the ideal maximum.

Part C: Past Exam Questions (Source-Tagged) (20 marks)

C1. No-forwarding vs forwarding pipeline (Source: exam24r.pdf / exam24px1.pdf, Q7a, adapted) (7 marks)
A short ARM sequence runs on a five-stage pipeline.

(a) Explain how the answer changes when the CPU has no forwarding/hazard detection.

(b) Explain how forwarding changes the hazard picture.

(c) Name one case where a stall is still required even with forwarding.

C2. Smith predictor accuracy (Source: 2023_Final_Exam.pdf, Q2D, adapted) (6 marks)
Explain why a 2-bit Smith predictor usually performs better than a 1-bit predictor on loops that are mostly taken and then exit once.

C3. Pipeline depth and register cost (Source: exam2025-main.pdf, Q7 / Q14, adapted) (7 marks)
Briefly explain the tradeoff between deeper pipelines and pipeline-register overhead. Why can a deeper pipeline improve clock frequency but still fail to improve overall performance proportionally?